

NUMERICAL ANALYSIS

PROBLEM SET 5

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1. PROBLEM 1

Derive the trapezoid integration scheme as an Adams-Moulton method with $m = 1$ step:

$$\frac{w_{j+1} - w_j}{h} = \frac{1}{2}[f(t_{j+1}, w_{j+1}) + f(t_j, w_j)]$$

to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with initial condition $y(a) = \alpha$. Also derive the associated error term. Let $w_j \approx y_j := y(t_j)$.

- (a) Use the Lagrange form of the polynomial interpolant to write

$$f(t, y(t)) = P_1(t) + R_1(t).$$

Integrate $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$ from $t = t_j$ to $t = t_j + 1$.

- (b) Set $w_{j+1} = w_j + \int_{t_j}^{t_{j+1}} f(t, y(t))dt$, approximating the 2nd term with $\int_{t_j}^{t_{j+1}} P_1(t)dt$, and derive the method above.
- (c) The error term $R_1(t) = \frac{f''(\xi, y(\xi))}{2}(t - t_{j+1})(t - t_j)$ for f (some $\xi \in (t_j, t_{j+1})$) can be integrated from $t = t_j$ to $t = t_{j+1}$ to get the error associated with the Adams-Moulton scheme:

$$\text{Error} = Mh^3 f''(\mu, y(\mu)).$$

Do this integration to find value of M .

- (d) What is the order of this one-step implicit method?

Solution. For the sake of convenience, we're going to set $g(t) = f(y, y(t)) = y'(t)$, and furthermore, we assume a uniform grid over which we will work. So set

$$t_j = a + jh, \quad h = \frac{b - a}{N}.$$

So we have that

$$y(t_{j+1} - y(t_j)) = \int_{t_j}^{t_{j+1}} g(t)dt.$$

- (a) So if we are to interpolate the function $g(t) = f(t, y(t))$ at the nodes t_{j+1} and t_j , then the degree-one Lagrange interpolant follows as $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$,

where

$$L_{1,0}(t) = \frac{t - t_j}{t_{j+1} - t_j} = \frac{t - t_j}{h},$$

and

$$L_{1,1}(t) = \frac{t - t_{j+1}}{t_j - t_{j+1}} = \frac{t_{j+1} - t}{h}.$$

Thus, we have that

$$f(t, y(t)) = P_1(t) + R_1(t)$$

in conjunction with

$$P_1(t) = \frac{t - t_j}{h} f(t_{j+1}, y_{j+1}) + \frac{t_{j+1} - t}{h} f(t_j, y_j).$$

Integrating over our interval $[t_j, t_{j+1}]$ yields

$$\int_{t_j}^{t_{j+1}} P_1(t) dt = f(t_{j+1}, y_{j+1}) \int_{t_j}^{t_{j+1}} \frac{t - t_j}{h} dt + f(t_j, y_j) \int_{t_j}^{t_{j+1}} \frac{t_{j+1} - t}{h} dt.$$

□

2. PROBLEM 2

Consider the initial value problem

$$y' = \frac{4t\sqrt{1+y^2}}{y}, y(0) = 1$$

for $0 \leq t \leq 5$.

Challenge. Show that $y(t) = \sqrt{(2t^2 + \sqrt{2})^2 - 1}$ is a solution to the initial value problem. Is it unique?

Numerically implement the following methods below, and run with enough values of step-size h to demonstrate that each of has convergence of global discretization error with rate $\mathcal{O}(h^2)$.

- (a) Optimal RK2 method
- (b) Two-step Adams-Bashforth
- (c) One-step Adams-Moulton method
- (d) Heun method (predictor-corrector)

Plot the error versus the step size on a log-log plot for all methods on the same axes to compare. Comment on the advantages and disadvantages of each.

Solution.

□

3. PROBLEM 3

Consider the linear multi-step method to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with the initial condition $y(a) = \alpha$:

$$w_{j+1} - \frac{18}{11}w_j + \frac{9}{11}w_{j-1} + \frac{2}{11}w_{j-2} = \frac{6}{11}hf(t_{j+1}, w_{j+1}).$$

- (a) Is the method stable? If so, is it strongly or weakly stable? Show your work by considering the roots of the characteristic polynomial, noting that $\lambda = 1$ is a root, and you can even factor and write $P(\lambda) = (\lambda - 1) * (\dots)$.
- (b) Is the method convergent?
- (c) Is the method consistent?

Solution.

□

4. PROBLEM 4

Use a built-in ODE solver to numerically integrate the Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(x - y) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz\end{aligned}$$

with the following values for coefficients $\sigma = 10, r = 28, b = 8/3$.

- (a) Using the initial condition $(x_0, y_0, z_0) = (5, 5, 5)$, solve the system for $0 \leq t \leq 10$ with a time step $h = 0.001$. Make a 2D plot of x versus z and a 3D plot of x versus y versus z .
- (b) Use a new initial condition $(\bar{x}_0, \bar{y}_0, \bar{z}_0) = (5, 5, 5)10^{-5}$, repeat the simulations in part (a) and the plots in part (b).
- (c) Plot the L_2 -norm of the difference in the solutions obtained using the different initial conditions in (a) and (b). Comment on your results.
- (d) Repeat parts (a) - (c) with different solvers and different step sizes. Comment on your results. You may see differences when using some solvers but not using others. Given the differences observed when using different methods, how can you tell what to expect for the exact solution?

Solution.

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